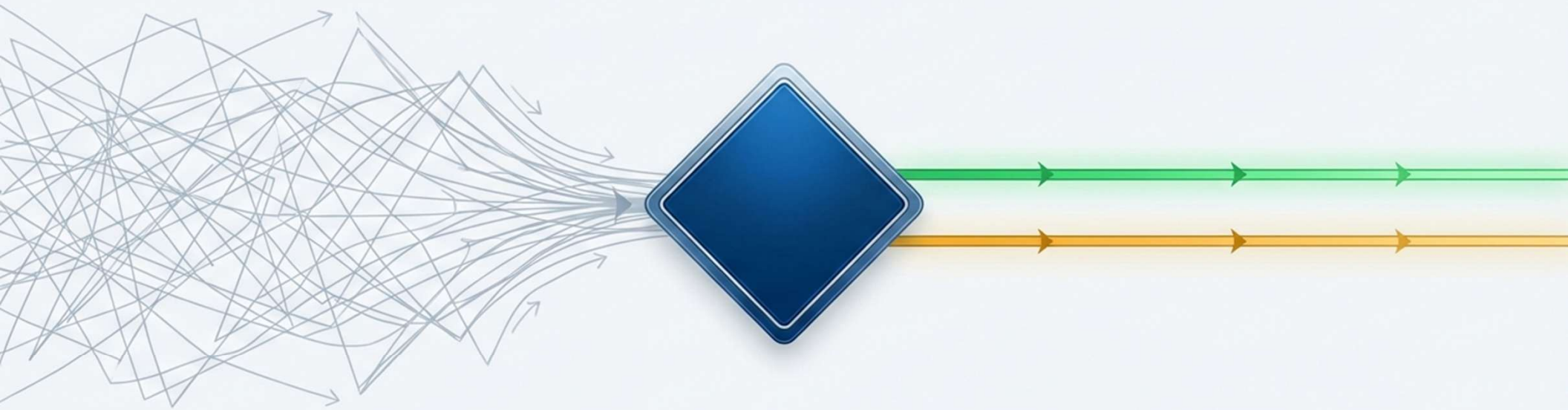


Automating the Signal in the Noise

Architecture, validation, and viability of the AI-Powered Career Opportunity Filtering System.



Developer: Tripti Kumari
Internship ID: 2026AICTECSRBOX1535

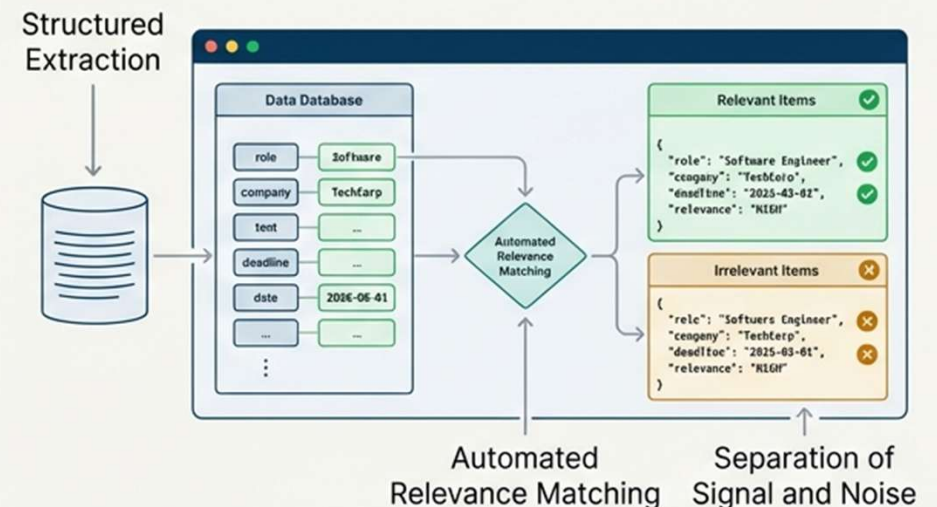
The Email Bottleneck in Early-Career Searches

The Status Quo



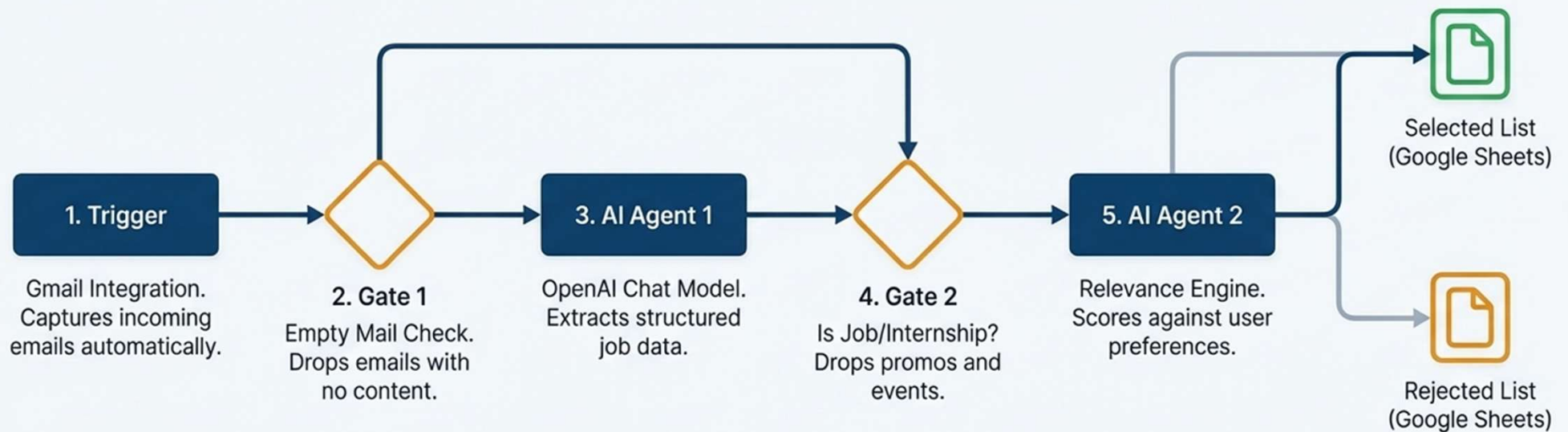
Outcome: High time consumption, frequent missed opportunities, decision fatigue.

The System Objective



Outcome: Zero manual effort, personalized filtering, actionable career intelligence.

A Dual-Agent Architecture for Precision Filtering



Agent 1: Strict Extraction and Classification

The Prompt Logic

Rule 1: Must have Hiring Intent + Role + Application Link.

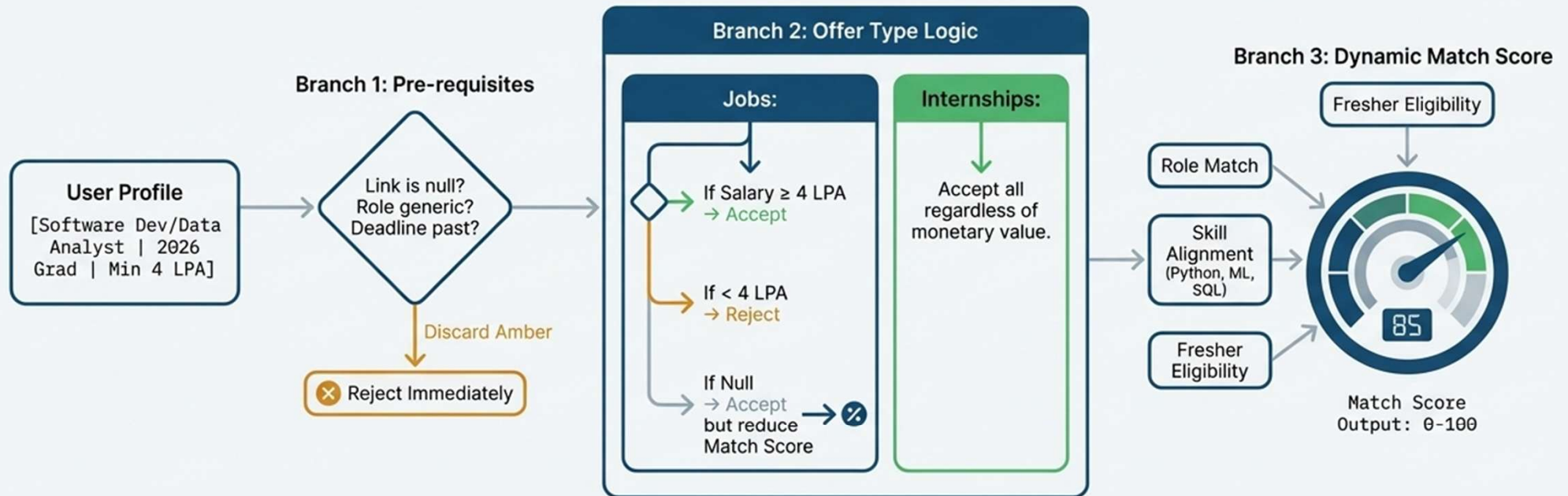
Rule 2: Ignore Newsletters, Promos, Events.

Rule 3: Extract ONLY top 1-2 relevant roles. Do not guess missing fields (Return Null).

```
{  
  "isJobOrInternship": "boolean",  
  "company": "string",  
  "role": "string",  
  "salary": "format: 'X LPA' or '₹X/month'",  
  "deadline": "YYYY-MM-DD"  
}
```

Strict JSON
Output Parsed
Automatically

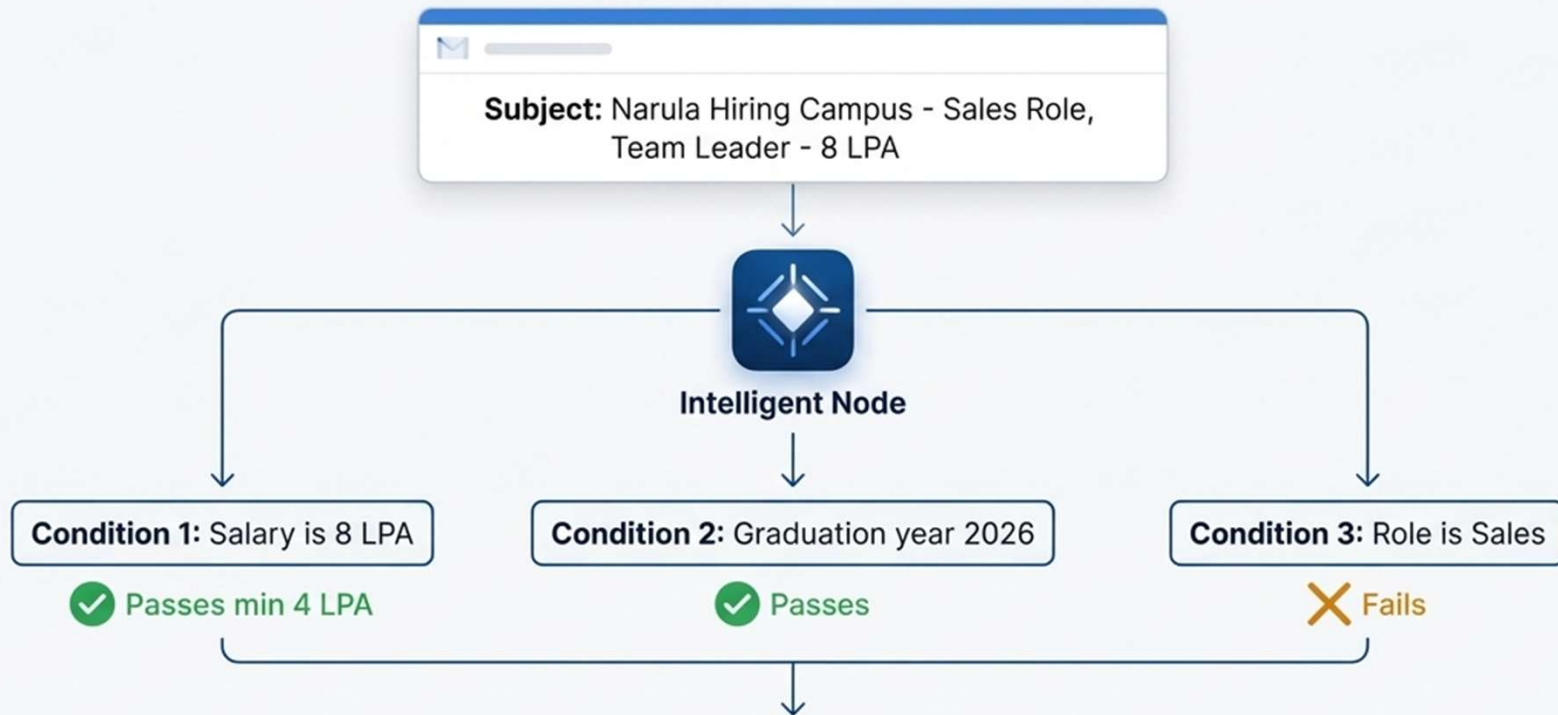
Agent 2: The Relevance and Scoring Engine



System Diagnostic Matrix: Edge Case Outcomes

Edge Case	Input Scenario	AI Decision Logic	Final Outcome
Case 1	Empty / No Content Email	Validates empty content.	Workflow Stops Prevents unnecessary API calls.
Case 2	Invitation to a party as guest speaker.	<code>isJobOrInternship = false</code>	Email Ignored Filters promotional noise.
Case 3	Sales role, 8 LPA.	Evaluates profile mismatch.	Rejected High salary but wrong domain.
Case 4	Genpact Data Analyst walk-in.	Identifies exact skill overlap.	Selected High match score.

Anatomy of a Smart Rejection (Edge Case 3)



isRelevant: false | Match_score: 30 | Routed to Rejected Sheet

Anatomy of a Precision Match (Edge Case 4)

Top Tier



Subject: Genpact - Kolkata Walk-In for Transaction Risk Investigator

Center Tier



Extracted Requirement:
Fraud detection, ML, SQL,
Power BI.



User Profile Map:
Prefers Data Analyst roles
(Direct Overlap).



Salary Check:
"Not Disclosed"
(Proceeds, flags for user).

Bottom Tier

> "Role involves transaction risk/fraud investigation which aligns well with candidate's hands-on data analytics experience. Caveat: Salary is not disclosed—confirm compensation before applying."

isRelevant: true | Match_score: 80 | Routed to Selected Sheet

The Output Layer: Structured and Actionable

Sheet 1 - Selected								
Sheet 2 - Rejected								
Success Green								
Date	Platform	Company	OfferType	Role	Salary	Deadline	Match_score	Summary
4/8/2026	Other	XYZ Analytics	Internship	Data Analyst Intern	₹15000/month	2026-05-10	95	Good fit: Data Analyst role aligns with user's preferences and strong skills (Python, SQL, Power BI, ML); suitable for a 2026 graduate and internships are accepted regardless of stipend.
4/8/2026	Naukari.com	Genpact	Job	Transaction Risk Investigator	Not Disclosed	2026-04-08	80	Relevant: The role involves transaction risk/fraud investigation which aligns well with the candidate's hands-on fraud-detection and data-analytics experience (built fraud detection on 6.3M transactions, ML, SQL, Power BI). Graduation year (2026) fits the job timeline and the candidate prefers Data Analyst roles, which overlap with this position. Caveat: salary is not disclosed (candidate's minimum is 4 LPA) — confirm compensation before applying.

Discard Amber			
Salary	Deadline	Match_score	Summary
			Role is a sales/team-leader position which does not match user's preferred roles (Software Developer, Data Analyst) or technical skillset. Although salary (8 LPA) exceeds the user's minimum and graduation year (2026) makes them eligible, the domain mismatch makes this opportunity not relevant. The listing is specific and includes an application link (not a generic email).
			Role not specified/unclear (reject). Internship is suitable for a 2028 grad but salary (~₹1.8 LPA) is below the user's 4 LPA preference.

Data is instantly available for tracking, filtering, and future career analysis without a single manual keystroke.

4/8/2026	Narula Hiring Campus	Job	Narula	sales role, team leader	8LPA	30	
4/8/2026	Other	Internship	XYZ Analytics		₹15,000/month	25	

System Economics & Deployment Strategy

Key Metrics



Number of emails processed automatically



Accuracy of job detection



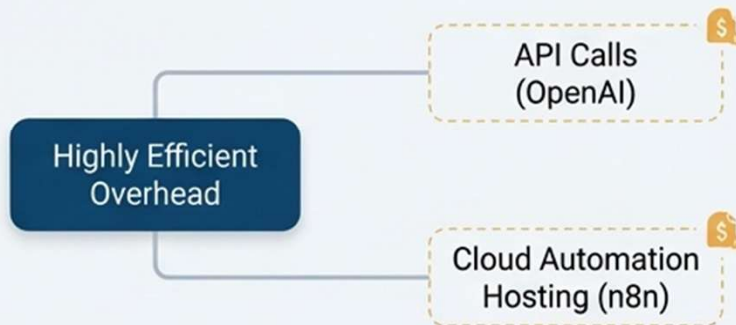
Reduction in manual hours spent per user

Unfair Advantage

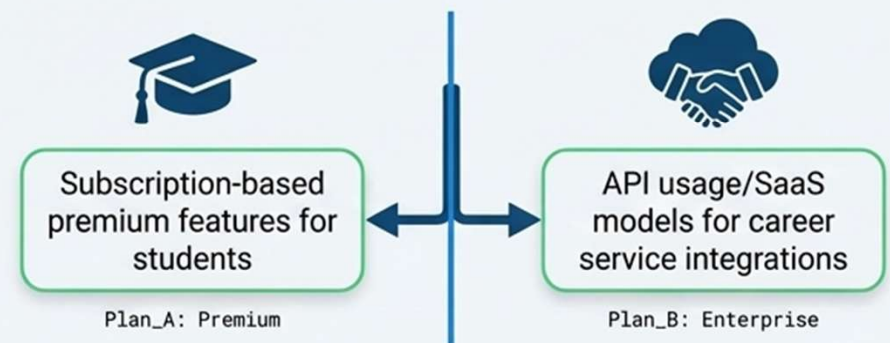


Custom AI workflow with personalized decision logic.
Real-time integration of extraction, filtering, and Google Sheets storage.

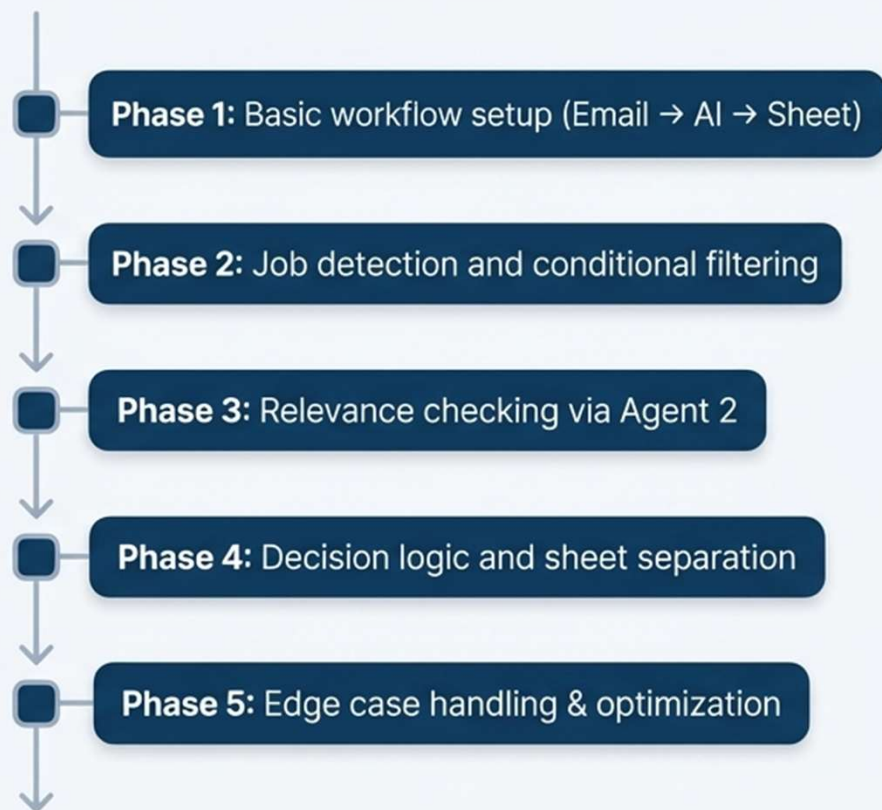
Cost Structure



Revenue Model



Project Roadmap and Resource Efficiency



Budget Breakdown (Estimated Monthly)

OpenAI API Usage: ₹1,000 - ₹2,000

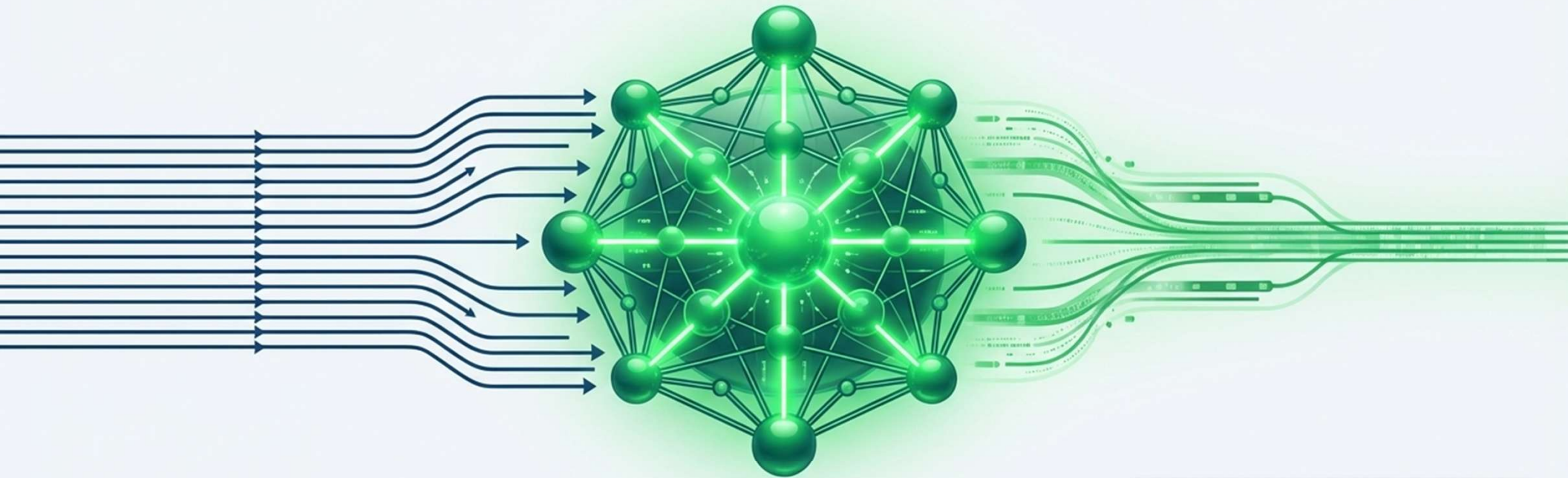
Automation Hosting (n8n): ₹0 (Local)

Data Storage (Sheets): Free

Total System Cost: ₹1,000 - ₹2,500

A highly scalable and
low-overhead solution.

Transforming Noise into Opportunity



By integrating dual AI agents with conditional workflow automation, we turn unstructured inbox chaos into a personalized, highly structured career advantage.

```
> Architect: Tripti Kumari  
> Program: B.Tech Computer Science Engineering  
> Institution: Narula Institute of Technology  
> Internship_ID: 202BAICTECSRBOX153S  
> Contact: vishwakarmap1385@gmail.com  
> LinkedIn_Status: Connected
```